



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO THARUN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

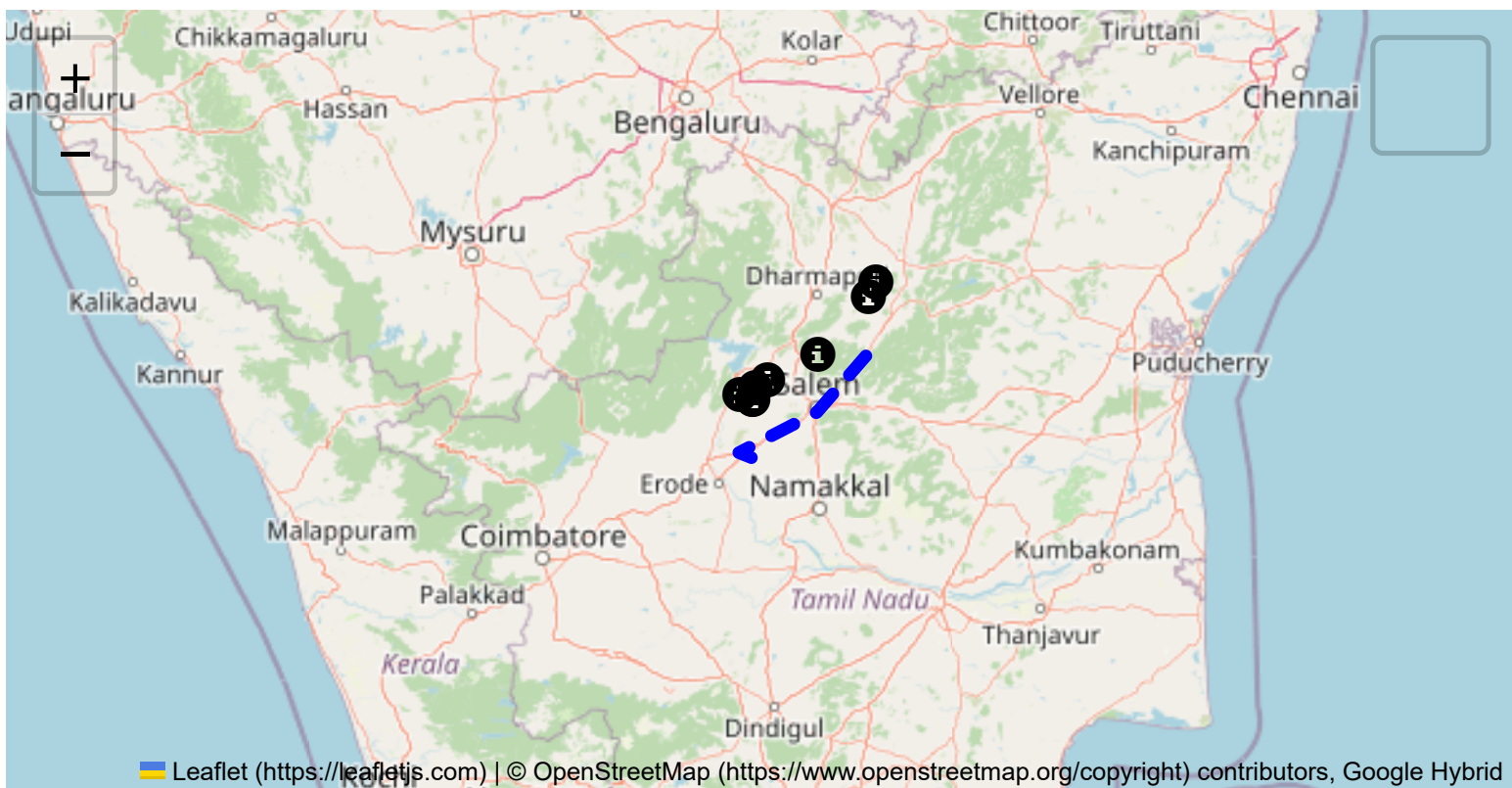
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

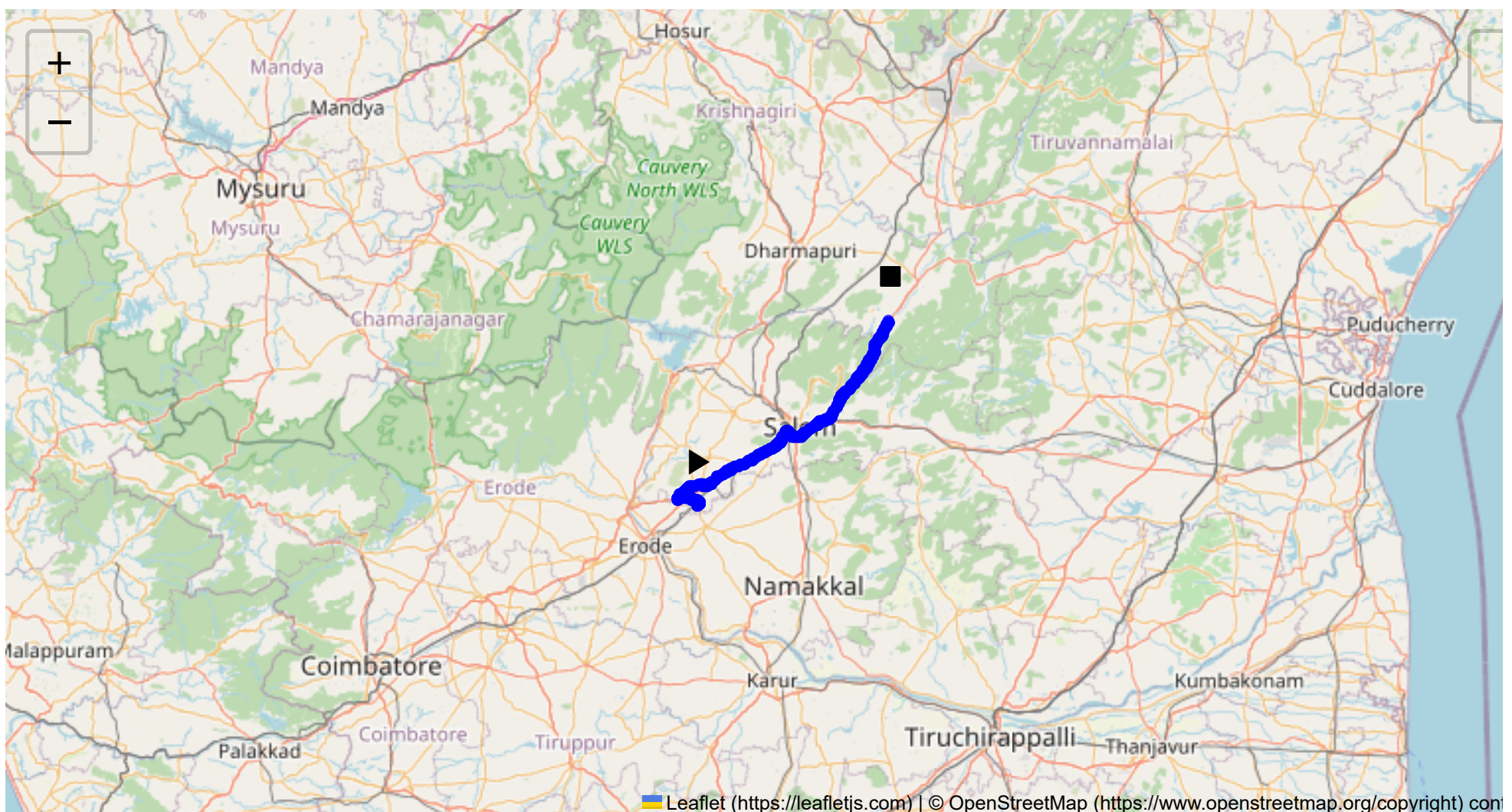
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 101.43 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (11.94202, 78.40631)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pallipatti is approximately 101.43 kilometers, taking about 2.43 hours for heavy vehicles carrying hazardous materials. Key stops along the route include Sangagiri, Puthur, Padaiveedu, Salem, Vaikundam, K R Nagar, and concluding in Pallipatti. The route involves various highways, including Bhavani Main Road, Salem-Kochi Highway, and NH 179A.

Weather Conditions and Hazards

- **Typical Weather:** Tamil Nadu generally experiences tropical weather with hot summers, moderate monsoons, and mild winters.
- **Potential Hazards:**
 - During the monsoon season (June to September), heavy rainfall could lead to slippery roads and localized flooding, affecting driving conditions.
 - In summer, high temperatures could lead to vehicle overheating.

Traffic Patterns and Congestion

- **Peak Hours:** Expect congestion during morning (7-10 AM) and evening (5-8 PM) rush hours, especially around Salem, which is a significant urban center.
- **Congestion Zones:** Salem and its surrounding areas, such as K R Nagar, are prone to higher traffic due to urban activities and commercial traffic.

Road Quality and Infrastructure

- **General Assessment:** The road quality varies, with major highways like Salem-Kochi Highway being well-maintained. Secondary roads may have potholes and uneven surfaces.
- **Infrastructure:** Pay attention to narrow stretches and two-lane roads that might be challenging for large trucks, especially in rural areas.

Alternative Routes for Emergencies

- If road conditions are poor, consider taking the NH 544 towards Coimbatore and then diverting to local roads leading to Pallipatti. However, ensure compliance with local transportation regulations for hazardous materials.

Local Regulations

- Adherence to Tamil Nadu Motor Vehicles Rules is essential, with specific permits required for transporting hazardous materials.
- Check legal restrictions on transport times and designated zones that either restrict or regulate heavy vehicle passage.

Historical Incidents

- Historical data on accidents involving heavy vehicles in this region suggests the importance of cautious driving near Salem due to high traffic density.
- Records of incidents often cite driver fatigue and poor vehicle maintenance as contributing factors.

Environmental Considerations

- The region has agricultural land and residential areas sensitive to hazardous material passage.

- Ensure correct handling and documentation to prevent environmental contamination and adhere to local safety standards.

Communication Coverage

- Communication signals are generally strong near urban areas. However, rural stretches might experience patchy mobile and data service.
- Carry alternative communication devices like a satellite phone or two-way radio.

Emergency Response Times

- **Urban Areas:** 15-30 minutes in and around Salem.
- **Rural Areas:** Could extend to over an hour depending on the proximity to major towns and availability of emergency services.

Overall Risk Assessment Summary

The route poses moderate risks due to potential weather impacts, congestion in urban areas, and varying road conditions. It is critical for transporters to adhere to local regulations, prepare for adverse weather, maintain communication tools, and continuously assess road conditions and traffic patterns. Emergency navigation plans and thorough pre-trip inspections are recommended to mitigate risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43961, 77.87341	15 KM/Hr
2	Turn	High	11.44024, 77.87528	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.45898, 77.85135	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	Medium	11.64158, 78.11808	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.64210, 78.11860	30 KM/Hr
13	Turn	Medium	11.64200, 78.12120	30 KM/Hr
14	Turn	Medium	11.64172, 78.12154	30 KM/Hr
15	Turn	Medium	11.64098, 78.12163	30 KM/Hr
16	Turn	High	11.63972, 78.12083	15 KM/Hr
17	Turn	Medium	11.62826, 78.13337	30 KM/Hr
18	Turn	Medium	11.65629, 78.20217	30 KM/Hr
19	Turn	High	11.66372, 78.20649	15 KM/Hr
20	Turn	Medium	11.67124, 78.22961	30 KM/Hr
21	Turn	High	11.67111, 78.23134	15 KM/Hr
22	Turn	High	11.67187, 78.23142	15 KM/Hr
23	Turn	Medium	11.67427, 78.23892	30 KM/Hr
24	Turn	Medium	11.68018, 78.24251	30 KM/Hr
25	Turn	Medium	11.81208, 78.34071	30 KM/Hr
26	Turn	Medium	11.83518, 78.35305	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
16	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
17	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
18	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
19	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
20	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
21	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
22	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
24	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
27	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
28	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
8	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
26	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64158, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64210, 78.11860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64200, 78.12120



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64172, 78.12154



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64098, 78.12163



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63972, 78.12083



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62826, 78.13337



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65629, 78.20217



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66372, 78.20649



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67124, 78.22961



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67111, 78.23134



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67187, 78.23142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67427, 78.23892



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68018, 78.24251



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305
